

**IX Open Autumn University Students' Mathematical Competition
OSAM Comp 2026 by FCS HSE**

September 13th 2026, 11:00 am – 2:00 pm

Category II+ (2nd year bachelors and further)

1. Let a and b be real numbers such that $a < b$ and $ab > 0$. Let $f: [a, b] \rightarrow [a, b]$ be a continuous function. Prove that there exists a real number $x \in [a, b]$ such that $xf(x) = ab$.
2. Does there exist a nonconstant polynomial $P(x)$ with real coefficients such that for every positive integer n the number $P(n)$ is a term of the Fibonacci sequence? Note. The Fibonacci sequence $\{F_1, F_2, \dots\}$ is defined by $F_1 = F_2 = 1$ and $F_{n+2} = F_{n+1} + F_n$ for $n \geq 1$.
3. Let A_0 be an invertible real symmetric $d \times d$ matrix. Define the sequence

$$A_{n+1} = \frac{1}{2} (A_n + A_n^{-1}), \quad n \geq 0.$$

Prove that all A_n are well defined and investigate whether this matrix sequence necessarily converges.

4. Let V be a real vector space of dimension at least 3, fix a nonzero vector $u \in V$, and let $A: V \rightarrow V$ be a linear operator. Suppose that $A(x) \in \text{span}\{x, u\}$ for every $x \in V$. Prove that there exist a scalar λ and a linear functional φ on V such that

$$A(x) = \lambda x + \varphi(x)u.$$

5. Let $a_1, a_2, \dots, a_k$ be distinct integers, with $\sum_{i=1}^k a_i = n$, and let g be the permutation

$$g = (1, 2, \dots, a_1)(a_1 + 1, a_1 + 2, \dots, a_1 + a_2) \cdots \\ \cdot (a_1 + \cdots + a_{k-1} + 1, \dots, n).$$

Prove that the centralizer of g in Sym_n , the symmetric group on n elements, is an abelian group.

Recall: the centralizer of g in Sym_n is the subgroup of permutations in Sym_n that commute with g .